

Hearing the Way Forward: Exploring Ambient Navigational Awareness with Reduced Cognitive Load through Spatial Audio-AR

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ABSTRACT

Notifications and alerts can both convey critical data needed to make decisions as we go about our day, and at the same time be completely ignored as a result of the overuse of these channels. Auditory notifications can be especially intrusive given their demand for synchronous attention. However, during some activities, such as navigation guidance via GPS, immediate information cues are necessary. In this case audio alerts and instructions are typically implemented as voice commands, but processing language requires significant cognitive resources and attention. In this study we report findings about the feasibility of a novel ambient pedestrian guidance system inspired by neuroscience. Results suggest that it is possible to communicate precise spatial actions without the intrusion of verbal turn-by-turn cues. We also share learnings from our use of low-fidelity prototyping to evaluate the novel audio-AR interaction.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**; **Interaction design process and methods**; **Interface design prototyping**; *User centered design*.

KEYWORDS

spatial audio, pedestrian navigation, augmented reality, planning, wayfinding

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1 INTRODUCTION

Notifications and alerts generated by our usage of mobile devices can help to bring our attention to contextually relevant information, time sensitive events, and critical data needed to make decisions as we go about our day. However, competing interests to guide our attention have resulted in an overuse of these channels that has devolved into a nearly continuous stream of sensory distraction, the psychological effects of which are still being investigated. It is known, however, that task switching imposes a considerable drain on cognitive resources [12]. Auditory notifications can be especially intrusive given their demand for synchronous attention. However, there are a handful of activities for which synchronous, immediate information cues provide critical value to users. One class of these activities is navigation guidance via GPS systems, where verbal cues help us decide where to turn to efficiently reach our destination. Even in these activities, however, audio alerts and instructions, which are typically implemented as voice commands, act to disengage us from our environment, at least momentarily, in order to process and subsequently act on the information provided. Given the wide diversity of navigation contexts, such disruptions can be especially problematic, e.g. when carrying on a conversation with a friend while trying to navigate to a nearby café.

Ambient information systems have been proposed to relieve this cognitive burden through low-attention peripheral interfaces. According to the taxonomy proposed by Pousman and Stasko, ambient information systems can be understood to vary on four dimensions: information capacity, notification level, representational fidelity, and aesthetic emphasis [13]. Prior work has demonstrated the efficacy of spatial audio cues to support pedestrian navigation [4, 8]. These studies have shown that a single localized audio beacon can effectively guide pedestrians wearing stereo headphones to a predetermined target location. However, the communication of richer multi-dimensional information, and the representational fidelity achievable in an ambient information system, remains an open question. In the present study, we build on this prior work, and take inspiration from several results from the neuroscience of navigation, as well as Monsell's finding that task switching costs can be reduced by an opportunity for preparation [12].

Spatial Awareness through Navigational Preplay. We hypothesized that communicating upcoming navigation decisions could enable pedestrians to efficiently move toward their destination in a

way that is significantly less distracting than traditional turn-by-turn guidance. While turn-by-turn commands are optimized for a reactive, synchronous, environmentally detached response, processing the next few actions in advance might enable the creation and recall of a more effective internal mental map of the environment. If such information can be conveyed through ambient, non-verbal, and spatially grounded audio cues, perhaps a balance of informational richness, and non-intrusion can be achieved. To explore the informational capacity and representational fidelity (and therefore feasibility) of such a design, we mocked up a web-based prototype of the augmented-reality (AR) experience using a 360-degree street view as visual stimulus, and evaluated users' ability to accurately interpret our non-verbal navigation instructions.

1.1 Prior Work

The literature on low-attention, hands-free, heads-up pedestrian navigation aids has explored a wide variety of design approaches. We review several of the most promising, which informed the present work, here.

1.1.1 Spatial audio & beacon navigation. When combined with a head-direction aware system, spatial audio generation through head-related transfer functions (HRTF), can produce a world-locked and immersive 3D audio experience. One of the most compelling findings relevant to this work, observed in a variety of prior studies, is that a single beacon producing intermittent spatialized audio originating at the destination can provide an unobtrusive and effective navigation guidance signal. A 2006 study introduced the SWAN system, which used HRTF-generated spatial audio to place beacons at each location in a series of waypoints guiding users through each step of a walking path [16]. Both sound design (broader-range sounds were easier to localize) and a mid-sized capture radius (defining completion of a segment) affected participant's performance. After 1-2 trials, most participants learned to follow the specified path with over 90% efficiency. More recent work used spatial audio beacons to guide pedestrians through a "scavenger hunt" of waypoints [3, 9]. This study found that spatial understanding (as measured by accurate identification of POI locations on a map following the navigation task) was significantly improved for a group using the beacon compared with a group using verbal turn-by-turn audio cues. Elaborating on the use of intermittent audio beacons for navigation guidance, Albrecht et al. explored the use of spatialized music as a navigation guide, comparing several guidance styles [1]. A number of findings from this study inform the present work. First, a significant limitation was identified in route guidance when users approached either a) a fork with an acute angle differentiating the two options, and b) a path segment with two nearby turns. In both cases, route guidance was ambiguous about which turn to take. The authors also found that beacon guidance could in some cases lead to dead ends. Finally, some participants indicated a desire to query for more information at moments of uncertainty: "I would like to be able to ask for directions in unclear situations." (p. 16).

1.1.2 Direction-aware guidance. The head-direction-awareness that enables world-locked spatial audio also opens up the design space to contextual real-time interactions capable of producing feedback based on the user's present orientation, and therefore, approximate

direction of gaze. Providing route guidance, especially to reduce ambiguity at decision points, has been explored in a number of studies. In RunAhead, Gallo et al. leveraged natural head scanning behavior to provide audio-based feedback to runners in order to guide choices upon approach to a decision point [6]. This targeted feedback was provided automatically when within a capture radius of an intersection, however by modulating feedback as runners scanned through visible choice options, this design offered a user-controlled query mechanism, and avoided providing less useful information (e.g. information about options the runner was not considering). Giannopoulos et al. added further precision to the direction-based guidance of prior studies by eye-tracking participants and providing vibrotactile feedback during fixations on a segment of the single optimal route to a destination [7].

1.1.3 Neuroscience of Navigation. A long scholarly history studying rodents has hinted at the neural underpinnings enabling navigation through complex environments. In the 1930s, Edward Tolman documented a somewhat familiar behavior: when rats came across an unknown fork in the maze, they would shift their head down one path, then the other, seeming to deliberate on the best option. Tolman dubbed this behavior vicarious trial and error (VTE) [15], and found that the incidence of VTE declined as rats learned the structure of a given maze.

More recently, the discovery of place cells in the hippocampus, as well as a variety of cell types in nearby medial entorhinal cortex, has begun to explain, in surprising nuance, the specific representational components available to the brain to support reasoning about where an organism is, what is nearby, and what it needs to do to get to where it is going. Head direction cells, for example, egocentrically code for the head's orientation [14], while object vector cells represent a relative and allocentric location defined with respect to certain types of obstacles in the environment [10].

Fascinatingly, these location-coding cells in the hippocampus appear not only to code for an animal's present location, but also predictions of where it may soon be. This prospective coding has since been linked with the VTE observed by Tolman [11].

The specific dynamics by which animals leverage these simulated forecasts down a candidate path is not well understood. However, one beneficial feature has been studied—the ability to run such simulations faster than they could be performed in reality. This time compression has been observed to extend from 2-4 times real-world velocity, and may enable experimentation, evaluation, or preparation for a navigational choice prior to committing to it [2]. In this study, we take inspiration from these forecasting dynamics to design more natural, effortless navigation experiences.

1.2 Motivation

The present study contributes to a growing HCI literature exploring the efficacy of various non-visual modes to provide hands-free notifications for navigation support to pedestrians. Inspired by key ideas from the neuroscience of navigation—vicarious trial and error, and hippocampal preplay—we conceived and evaluated two novel designs to preview the next few segments of the route to a specified destination. We aimed to address three objectives in the proposed designs: 1) to explore a non-verbal way of delivering navigational

guidance, 2) to convey future action information enabling a continuous, easily anticipated navigation experience, and 3) to ground route information and improve understanding of spatial structure through embodied interaction with the real world.

While this study was largely exploratory—investigating the design space and evaluating the feasibility of novel route guidance designs—we sought to validate several hypotheses about participants’ performance on the task, and its relation with mode and complexity. In particular, we predicted that:

- (1) Due to the cognitive demands and novelty of the interaction, performance would be lower in simulated audio-visual trials versus isolated audio trials¹.
- (2) Performance would be lower in more complex paths (with more instructions, and more turns).
- (3) Angular destination distance would be lower in simulated trials. We expected the world-locked nature of the interaction to provide a useful “heading to destination” cue which would reduce the angular offset.

In the following sections, we present the design of our spatial-audio preplay guidance system, describe the methods used for the feasibility study, and the analyses performed. We then present key results from our analysis, including comparisons of selected path accuracy metrics, and learning effects observed through the experiment. We end by discussing the implications of these results, as well as our main learnings from the design process including our use of neuroscience-inspired principles for ambient information design, and the use of a low-fidelity web-based AR prototype. Finally, we suggest a couple of avenues implicated for exploration of our design, and ambient navigation systems in general.

2 METHODS

2.1 Preplay Design

We designed two modes of non-verbal audio guidance to test in the feasibility study. **Isolated audio** prompts play a spatialized audio sequence representative of the steps (forward motion, and egocentric turns) in the indicated route. This mode used a vocabulary of short audio symbols to convey each movement. A set of grouped tones (single, double, or triple) played in a single ear (fully power-panned) to indicate either a slight turn, normal turn, or sharp turn to the left or right. A “pop” sound indicated movement straight ahead to the next intersection. Finally, a chime indicated the conclusion of the route².

Our second mode, composed of **simulated audio-visual** prompts, built on the same audio vocabulary, but added two significant extensions to enhance the spatial groundedness of the route information. First, “move straight” commands were localized using spatial audio such that they originated from the position along the route in real-world coordinates. In this way, users were able to hear a “leader” as it moved through the real-world route, emanating sounds as it traveled. Second, this mode used a more embodied, interactive design in which participants tracked the locations of key waypoints (e.g. turns in the route) by orienting their head towards each waypoint’s

location. We hoped that the side-to-side scanning movement, similar to that observed in vicarious trial and error, might support the development of a spatially realistic map of the upcoming path segments.

To create a more faithful, embodied experience within the confines of an online study, we used a 360-degree photo sphere of the starting location, and allowed participants to control the direction of view using their mouse³ (see Figure 2). After orienting the view to look at the next waypoint, audio guidance for the subsequent path segment was played. In this way, participants traced the entire route by shifting their view towards each waypoint in series. At the end of the interaction, participants would be facing directly towards the location of the destination. Examples of the simulated audio-visual interaction can be viewed in the tutorial video⁴. Figure 1 shows a schematic of each mode for an example route.

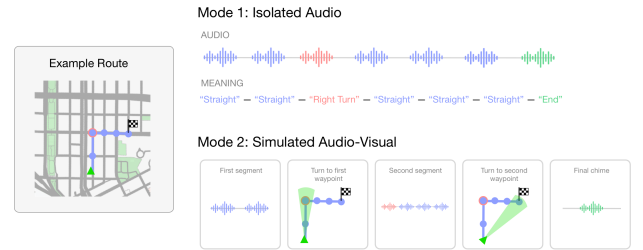


Figure 1: Schematic of two study modes. Isolated audio presented a symbolic audio sequence representing each turn using non-verbal cues. The simulated audio-visual mode allowed the participant to interact by turning their field of view towards each waypoint sequentially.

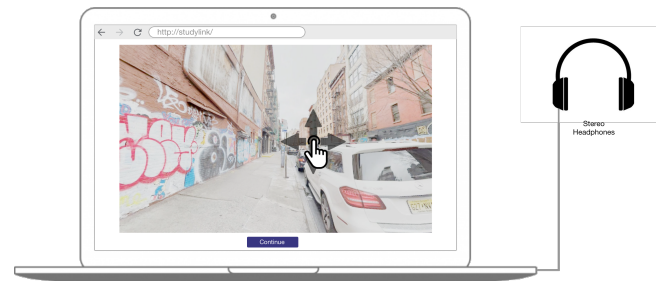


Figure 2: Schematic of web-based study setup showing 360-degree view in simulated audio-visual mode.

2.2 Online Feasibility Study

To validate the feasibility of communicating routes in each design modality (isolated audio, or simulated audio-visual), we ran an online experiment.

¹This prediction is primarily driven by the limited nature of evaluation in the context of our online study, and the difficulty to measure other potential benefits of this mode.

²Sample audio files can be found in the supplementary material

³The interaction was implemented using Three.js (<https://threejs.org/>) and react-three-fiber (<https://github.com/pmndrs/react-three-fiber>)

⁴Tutorial videos for both modes can be viewed in the supplementary material

2.2.1 Path Generation. We generated 16 paths distributed between four maps of urban locations. Locations were selected to be diverse in terms of navigational complexity (e.g. one map was on a grid-like portion of a down-town area, whereas another included an intersection entered on a 45-degree turn, and with 5 possible continuations). A single starting location (on a mid-block sidewalk) was chosen for each map, from which four paths were drawn. Paths varied in number of turns (0, 1 or 2), and number of instructions (3 to 10). Two example maps can be seen (with responses plotted) in Figure 4.

2.2.2 Experimental Design. To study performance across the two modalities, we used a mixed 2x2 design. Participants were randomized into one of two groups: those who saw the block of isolated audio prompts first, and those who saw the block of simulated audio-visual prompts first. Within each block, a random selection of 8 of the 16 paths was shown, with the block determining whether the route would be conveyed in isolated audio or simulated audio-visual form. This allowed aggregate comparisons between mode for a particular path, as well as between modes overall (since block order was counter balanced).

2.2.3 Study Protocol. Participants were recruited via email from an internal sample of employees at a US-based technology company, according to a protocol approved by an internal research ethics committee. Participants first completed a survey to ensure they met all inclusion criteria (e.g. the availability of stereo headphones, and the absence of any profound hearing or vision impairment). Qualifying participants were shown a consent form followed by a brief demographic survey. Next, participants were randomized into one of two groups, and watched a tutorial video introducing the overall study and how to interpret the information that would be presented in each prompt. A second tutorial video was shown prior to each block of the experiment, which provided examples of the prompts, as well as a visual guide illustrating the progression of the route instructions.

Next, each participant began the two blocks of the main study. For isolated prompts, a photo from the perspective of the start location was shown next to the audio player. For simulated prompts, a single large window contained the interaction, with a similar initial appearance looking down the road from the hypothetical start location. Each prompt could be listened to—or interacted with (in the case of the simulated mode)—as many times as needed. Once participants were ready, they clicked a button to move to the response screen. Responses were collected by showing a map of the region around the start location and asking participants to select the destination they believe was indicated by the prompt. This was done by clicking a sequence of links coinciding with each intersection participants believed the route passed through on the map. Feedback on accuracy was not provided. After completing the 8 prompts for the first block, the second introductory video was shown, followed by the second and final block of the 8 remaining prompts. Due to the internal sample, and short duration of the experiment, compensation was not offered for this study.

2.3 Data Analysis

Data collected from the online study included, for each prompt, response duration, number of replays, and the sequence of intersection IDs selected by the participant. The simplest metric (path accuracy) required only a comparison of the participant’s intersection ID sequence with the true sequence.

A number of accuracy metrics, however, required additional spatial and structural information about the map as well as the chosen and correct paths. To generate this additional context, we hand-coded the positional and graphical connectivity of each intersection. This allowed us to compute accuracy metrics (e.g. angular offset to destination) using simple trigonometry. Additionally, to support metrics such as move edit distance and turn type mismatch, we inferred the correct sequence of “moves” for each path by moving a cursor through the correct intersection sequence, and bucketing changes in heading (in degrees) $dH = |\theta_i - \theta_{i-1}|$ as follows: $dH < 22.5 \rightarrow$ “straight”, $dH \in [22.5, 67.5) \rightarrow$ “slight turn”, $dH \in [67.5, 112.5) \rightarrow$ “normal turn”, and $dH \in [112.5, 180] \rightarrow$ “sharp turn”. Direction of turn was determined by the sign of the heading change.

Edit distances for both moves, and paths, were computed using an unweighted Levenshtein distance of movement types, and intersection IDs, respectively. For a detailed description of each metric, see below.

For the purposes of the variable definitions below, “response destination” refers to the last intersection selected by the respondent.

3 RESULTS

33 participants completed the study, reasonably balanced with respect to age: (25-34, 38%, 35-44, 35%, 45-54 18%, 55-64, 9%), and gender: (Male, 62%, Female, 32%, Non-Binary, 6%). 15 participants saw the isolated audio block first, and 18 saw the simulated audio-visual block first. Analysis of data collected from the online study included within subjects and between subjects comparisons validated by pairwise t-tests of each of our primary response accuracy metrics (described below).

Our main findings are summarized in Table 2. For a detailed description of each metric, see the Table 1.

In an analysis of the effect of block order on path accuracy, we find a significant interaction between block order and block type (ANOVA, $F(1, 31)=7.2$, $p=0.012$). The isolated audio block (when following the simulated audio-visual block) showed the highest path accuracy ($T=3.27$, Cohen $D=1.20$, $p=0.007$, see Figure 3).

In trials where participants failed to select the exact path indicated by the prompt, we can say that at least one error was made in interpreting the audio or audio-visual information. In an attempt to compare the types of errors made between modes, we observed individual trial data (e.g. via analysis of map traces like those shown in Figure 4) and noted common error types. We then systematically bucketed error trials based on the 3 types identified: overshoot by one (in which participants indicated a path that moved past the destination by one intersection), turn direction mismatch (e.g. left turn instead of right turn), and turn type mismatch (e.g. sharp turn instead of slight turn). See the frequency of each error type in Figure 6. Mode differences in error type frequency were significant according to a Chi-square test ($\chi^2 = 27.3$, $p < .0001$, $df = 4$).

Isolated audio resulted in statistically improved performance according to all path accuracy metrics except for two: straight count difference, and turn count difference.

We note, however, that performance in the second phase of each block was significantly different (between modes) only for graphical destination distance. This finding may indicate that, once learned, the two modes afforded comparable average accuracy.

Learning effects. Given the novelty of the instruction modes we tested, learning effects were expected. Furthermore, while each tutorial video included several sample prompts allowing participants to test their understanding of the prompt design prior to starting the study, the simulated mode, in particular, was best understood through experience. We observed two types of learning in this study, which we consider familiarity improvement and performance improvement.

The first is well illustrated by time series plots of trial duration (see Figure 7), which show clear downward trends that are especially steep in the simulated blocks. The novelty of a view-direction dependent interaction, controlling the field of view with a cursor, and listening for spatially localized audio cues, all required more

Table 1: Definitions of accuracy metrics used.

Metric	Definition
Valid	1 if response intersection sequence defines a single, connected path through the annotated graphical structure of the map, 0 otherwise.
Path Correct	1 if sequence of response intersections exactly matches coded solution sequence, 0 otherwise.
Destination Correct	1 if response destination matches true destination, 0 otherwise.
Path Edit Distance	Unweighted Levenshtein distance between response intersection sequence and solution sequence. Graphical structure is ignored.
Move Edit Distance	Unweighted Levenshtein distance between response action sequence and solution action sequence. Graphical structure is ignored. Actions are defined by the process described in the Data Analysis section of the Methods.
Graphical Destination Distance	Length of shortest path between response destination, and true destination.
Angular Destination Distance	Absolute value of difference between angle from origin to response destination, and angle from origin to true destination (degrees).
Straight Count Difference	Absolute value of the difference between the response sequence length and true sequence length (path length difference).
Turn Count Difference	Absolute value of the difference between the number of turns indicated by the response, compared with those indicated by the solution.

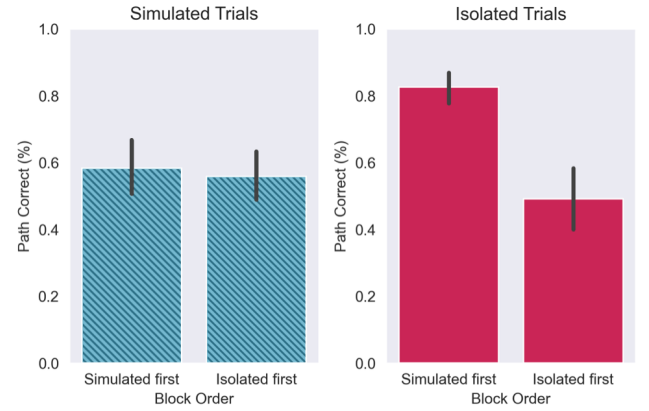


Figure 3: Plots showing the effect of block order on path correctness. Performance was significantly improved in isolated trials when they appeared second, following a block of simulated trials. The inverse dynamic is not observed, indicating an asymmetry in the transfer of learning between modes.

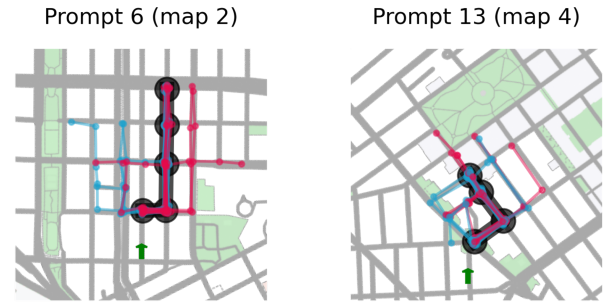


Figure 4: Plots of routes selected for two sample maps. Colors indicate mode (blue for simulated, red for isolated). Black outlines indicate the correct path. A spatial jitter is added to aid readability.

time for participants to understand and gain comfort with. The second learning effect was a general improvement in path accuracy as participants progressed through the experiment.

To evaluate something closer to a post-learning performance between modes, we additionally segmented our main analyses by “block phase,” separately grouping the first 6 and final 2 trials in each block of 8. To the extent that learning occurred over the course of a block, we expected performance in the final 2 trials to better represent potential performance on the task, once users had become familiar with the prompt design. Our main analyses, across all performance metrics studied, illustrate this expected improvement (see Table 2).

4 DISCUSSION

In this paper we present a new way of conveying rich, spatial information without the use of verbal cues, thus making it possible to provide ambient information to users while they are not actively

Table 2: Summary of primary accuracy metrics. Values reported are mean within mode and block “phase” (first 6 trials in block, or last 2 trials in block). Right column indicates results of independent sample t-test comparison of mode means, reported for $p < 0.05$. For a detailed description of each metric, see the Table 1.

Accuracy Metric	Isolated audio		Simulated audio-visual		Mode sig. diff.
	Early Phase	Late Phase	Early Phase	Late Phase	
Valid	.96 ± .01	.95 ± .03	.95 ± .02	.91 ± .04	-
Path Correct	.66 ± .03	.71 ± .06	.55 ± .04	.65 ± .06	*T=2.43, p=.015
Destination Correct	.69 ± .03	.74 ± .05	.58 ± .04	.68 ± .06	*T=2.29, p=.022
Graphical Destination Dist.	.69 ± .09	.55 ± .15	1.2 ± .13	.83 ± .18	*T=-3.72, p=.001
Path Edit Dist.	.19 ± .02	.15 ± .04	.29 ± .03	.21 ± .04	*T=-3.18, p=.002
Angular Destination Dist.	6.81 ± 1.19	4.67 ± 1.43	13.6 ± 1.68	9.11 ± 2.57	*T=-3.63, p<.001

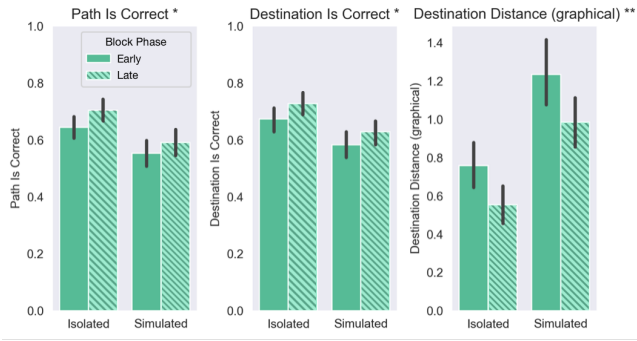


Figure 5: Comparison of path accuracy means segmented by mode and block phase. A single star in the title indicates significant difference between mode means ($p < 0.05$), and two stars indicates significant difference between mode means in the second phase of each block. Bars indicate standard error of the mean (SEM).



Figure 6: Left: Plot showing the relationship between the number of turns in a path, and path correctness. Participants performed significantly better on more complex 2-turn paths during the isolated audio mode. Right: Plot showing the rates of the most common error types. Turn direction errors were more common in simulated mode trials.

paying attention. We wish to highlight two main study features for which our learnings may be valuable to HCI practitioners. First,

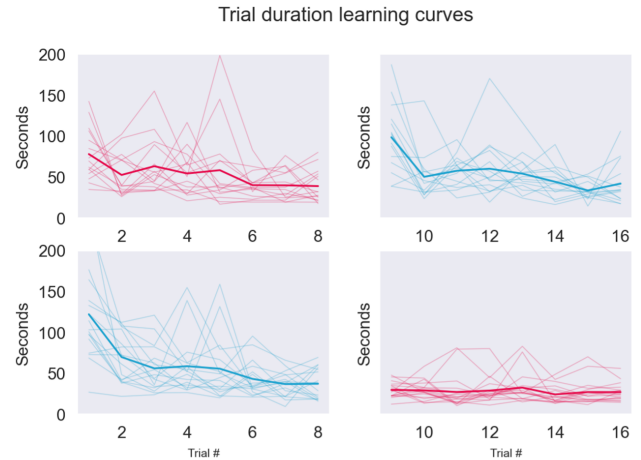


Figure 7: Time series plots showing learning curves according to trial duration. The top row shows participants seeing the isolated audio first (red), and the bottom row shows participants seeing the simulated audio-visual block first (blue). Two trial durations were over 3 standard deviations above the mean, and were removed from this analysis.

to explore the bounds of rich, yet ambient, information access, we developed a design inspired by concepts from the neuroscience of navigation. Second, to quickly generate feedback on our proposed designs, we employed a low-fidelity prototype of the novel audio-AR interaction which could be experienced, and therefore validated at scale, through a web browser and stereo headphones.

Neuro-Inspired Ambient Information. While spatial audio is able to confer intuitive directional guidance with surprisingly high precision, the upper limits of its information capacity have not been well characterized. Findings from neuroscience studies of navigation in both rodents, and humans—specifically, concepts known as vicarious trial and error, and mental preplay—are suggestive of the kinds of cognitive and behavioral dynamics at play during pedestrian wayfinding. Based on these concepts, we designed a richly embodied, interactive navigational guidance system capable of communicating multiple upcoming locations, each coupled with

a route instruction. By sweeping the head left and right to actively explore the path ahead, we hoped participants would gain a spatially grounded understanding of their position relative to their destination, along with key decision points immediately ahead.

While the efficacy, precision, and affinity for the proposed interaction will need to be evaluated in a field study, the preliminary feasibility results reported here suggest that it is possible to communicate precise spatial actions without the intrusion of verbal turn-by-turn cues.

Low-Fidelity AR Prototypes. As has been pointed out in prior work, prototyping for augmented reality poses unique challenges to researchers and designers [5]. We found that evaluating a web-based prototype early in concept development offered a number of benefits. A mature ecosystem of web-based interactive libraries made it possible to develop and internally test the initial prototype within a tight time frame. We found that the ability to deploy and share the interactive design also enabled earlier feedback from testers and key stakeholders. An additional unexpected benefit was also realized. The web-based evaluation enabled us to test an alternative, simpler, navigation guidance mode within the same study context, therefore eliminating potential confounds. The isolated audio mode required participants to listen to a short spatial audio track in the browser, which was trivial to compare with results from the simulated mode. Because the feasibility study was conducted online, participants could be recruited through internal email lists, and complete the study on their own computer. This enabled far more completions than would have been possible with an in-person protocol.

The primary limitation of our chosen methodology is the one shared with all user feedback on a low-fidelity prototype—how well will findings generalize to real-world use? While the web-based AR interaction conveyed a similar visual experience and feel to what would be experienced by a pedestrian testing the design in a real city street, several divergences were unavoidable. Most notably, the limited field of view imposed by the computer screen, and disembodied nature of rotating the perspective made it much easier to lose track of one’s orientation (e.g. to forget one’s initial direction). This is a possible explanation for the increased “turn direction” error rate in the simulated mode. Additionally, real-world use of the audio guidance would involve variation in the state of the user when requesting guidance (e.g. they may be walking, or still, with head oriented in the direction of travel or not). The web prototype removed this variable and conveyed a view from a static position, with field of view aligned with the present street. Due to this simplification, challenges or design limitations which arise for a user in other states could not be explored.

5 CONCLUSION & FUTURE WORK

In this study, we explored an ambient information system that could replace verbal notifications of GPS using a novel encoding of rich multi-dimensional information (the locations of key upcoming route decision-points, and the associated actions at each), and evaluated its representational fidelity by testing a low-fidelity prototype of the audio-AR interaction. We found that participants were able to understand the non-verbal audio cues with sufficient accuracy

to accurately identify the exact path to destination in the majority of trials, after a relatively short learning period.

We find this result encouraging, and believe it supports continued exploration in this space. Specifically, future work should validate the presented feasibility results with a usability assessment of an audio-AR prototype implementing a preplay feature. Many open design questions remain, e.g., how and when to trigger audio guidance to non-intrusively support pedestrian navigation. Finally, we wish to note that the design space we began to explore in this work is vast, and invites additional evaluations of non-verbal spatial audio designs that may reduce the distraction and cognitive load of traditional navigation guidance systems.

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